DATASCI 406 FA 26, Quiz 2.1

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

For a random variable X and a probability p, what does the quantile function $Q_X(p)$ return?

1. The probability that X is less than or equal to p.
2. The smallest value x such that the CDF evaluated at x is at least p.
3. The value x at which the density of X equals p.
4. The value of the CDF evaluated at X.

Solution

Answer: 2. This is $Q_X(p) = \inf\{x : F_X(x) \geq p\}$ read aloud. Option 1 is the CDF evaluated at p, and option 3 confuses the density with the CDF. Option 4 is the probability integral transform $F_X(X)$, which is a $U(0,1)$ random variable rather than a value of x.

Question 2 (1 point)

X has CDF $F(x) = x^3$ for $0 \leq x \leq 1$. What is its quantile function $Q(u)$ for $0 \leq u \leq 1$?

1. u^3 2. $3u^2$ 3. $1 - u^{1/3}$ 4. $u^{1/3}$

Solution

Answer: 4. F is continuous and strictly increasing, so solve $u = x^3$ for x. Option 1 is F again, and option 2 is the density $f(x) = 3x^2$.

Question 3 (1 point)

Which line of R produces 1000 draws from an exponential distribution with rate 2, using `qexp()` and not `rexp()`?

1. `runif(1000, qexp(0, rate = 2), qexp(1, rate = 2))`
2. `qexp(1000, rate = 2) * runif(1000)`
3. `qexp(runif(1000), rate = 2)`
4. `runif(qexp(1000, rate = 2))`

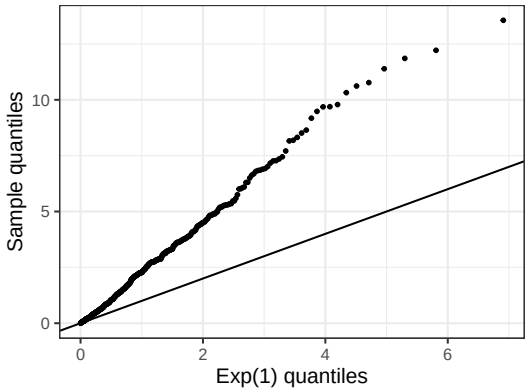
Solution

Answer: 3. Draw $U \sim U(0,1)$, then apply Q to it. Option 1 is uniform over the support (which here is unbounded, so it fails outright). Options 2 and 4 pass 1000 to `qexp()` as though it were a probability.

Question 4 (1 point)

A sampler is meant to draw from Exp(1). The QQ plot at right compares 500 of its draws with the Exp(1) quantiles, and the line is the 45 degree line. Is the sampler correct?

1. Correct: the points form a straight line.
2. Wrong: the points do not follow the 45 degree line.
3. Can't tell: a plot is not a hypothesis test.



Solution

Answer: 2. The points follow a line of slope about 2, not the 45 degree line. (Reading the direction, each sample quantile is about twice the matching Exp(1) quantile, but the quiz does not ask for that.) The sampler used `-2 * log(runif(n))`, which is Exp(1/2). A straight line means the right family at the wrong scale; only the 45 degree line means the right distribution.

Question 5 (1 point)

X takes the values below. To draw X you generate $u = 0.53$ and apply the inverse transform method for discrete variables. What value x do you generate?

x	10	20	30	40
$\Pr(X = x)$	0.2	0.3	0.1	0.4

1. 10 2. 20 3. 30 4. 40

Solution

Answer: 3. The running totals are 0.2, 0.5, 0.6, 1.0. The total first reaches 0.53 at $x = 30$, since $F(20) = 0.5 < 0.53 \leq 0.6 = F(30)$. Option 2 stops at the last total below u instead of the first one above it. Option 4 picks the largest probability, or the one closest to u, which skips the running total entirely.

Rubric. 1 pt per question, no partial credit.